Northeastern

Social Contagion - Rumor Model

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Laboratory for the
Modeling of Biological +
Socio-technical Systems

March 9, 2026

Epidemic Theory -> Social Contagion

No. 4955 October 17, 1964 NATURE 225

GENERALIZATION OF EPIDEMIC THEORY

AN APPLICATION TO THE TRANSMISSION OF IDEAS

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Table 1. ANALOGY BETWEEN INFECTIOUS DISEASE AND INTELLECTUAL EPIDEMICS

Elements of the epidemic process	Elements interpreted in terms of Infectious disease epidemic	Intellectual epidemic
Host		
Agent	Infectious material	Idea
Infective	Case of disease	Author of paper
Susceptible	Person who will be infected given effective contact	Reader of paper who will be infected given effective contact
Removal	Death or immunity	Death or loss of interest
Vector		
Agent	Infectious material (as for host)	Idea (as for host)
Infective	Vector harbouring the agent	Paper containing useful ideas
Susceptible	Vector not harbouring the agent	All papers containing potentially useful ideas
Removal	Death	Deletion or loss

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Rumor Model

I

Ignorant

Has not heard the rumor

S

Spreader

Actively spreading the rumor

R

Stifler

Knows the rumor, but is no longer spreading it

U R E December 12, 1964 VOL. 204

STATISTICS

Epidemics and Rumours

Goffman and Newill[1] have directed attention to the analogy between the spreading of an infectious disease and the dissemination of information. We have recently examined the spreading of a rumour from the point of view of mathematical epidemiology and wish to report very briefly here on work to be published in detail elsewhere[2]. In particular, we must emphasize that a mathematical model for the spreading of rumours can be constructed in a number of different ways, depending on the mechanism postulated to describe the growth and decay of the actual spreading process. In all these models the mathematical techniques familiar in mathematical epidemiology can be applied, but even the qualitative results so obtained need not necessarily be as expected on the basis of the formal analogy with epidemics.

To illustrate this point (but not to advance the model *per se*), let us divide the (finite, closed, homogeneously mixing) population into three mutually exclusive and exhaustive classes:

Class	No.	Rumour interpretation	Epidemic interpretation
X	x	Has not heard rumour	Susceptible to disease
Y	y	Actively spreading rumour	Infectious case
Z	z	No longer spreading rumour	Dead, isolated, or immune

If we assume that the only transitions are: (1) 'infections' $(x, y, z) \rightarrow (x-1, y+1, z)$, at a rate proportional to xy; (2) 'removals' $(x, y, z) \rightarrow (x, y-1, z+1)$, at a rate proportional to y; then we obtain the deterministic Kermack–McKendrick epidemic[3] in which an initial trace of infection will generate an appreciable epidemic if, and only if, the initial number $x(0)$ of susceptibles exceeds a certain 'threshold' value ρ. In this formulation the decay of spreading is due to a process of 'forgetting'. Another plausible hypothesis is that an active 'spreader' stops telling the rumour because he learns that it has lost its 'news value'; if this happens as soon as he meets another individual knowing the rumour (that is, a member of Z or another member of Y), then transitions from Y to Z occur as a result of YZ and YY encounters (with a double transition in the latter case), and hence at a rate proportional to $yz + 2[\frac{1}{2}y(y-1)] = y(y+z-1)$.

If this assumption is made, however, the behaviour of the system is strikingly different from that of a Kermack–McKendrick epidemic, and could not have been predicted from familiarity with the latter. The fraction f of the population which ultimately learns the rumour (corresponding to the fraction of the population which succumbs to the disease) is always about 0·80, almost irrespective of the value of $x(0)$, so that the familiar threshold effect has disappeared. A stochastic analysis yields very similar results; we find that f (now a random variable) has an expectation of about 0·80 and essentially a variance of approximately $0{\cdot}31/N$, where $x(0) = N$.

'Reluctance to tell stale news' can be incorporated into the model in other ways, some of which we have investigated by similar methods, and the results are then correspondingly modified. However, the object of this communication is not to urge the appropriateness of any one model of rumour-telling; but to stress the danger of transferring formulæ designed for epidemiological use without a thorough re-examination of the hypotheses, which may not always be appropriate in the new situation.

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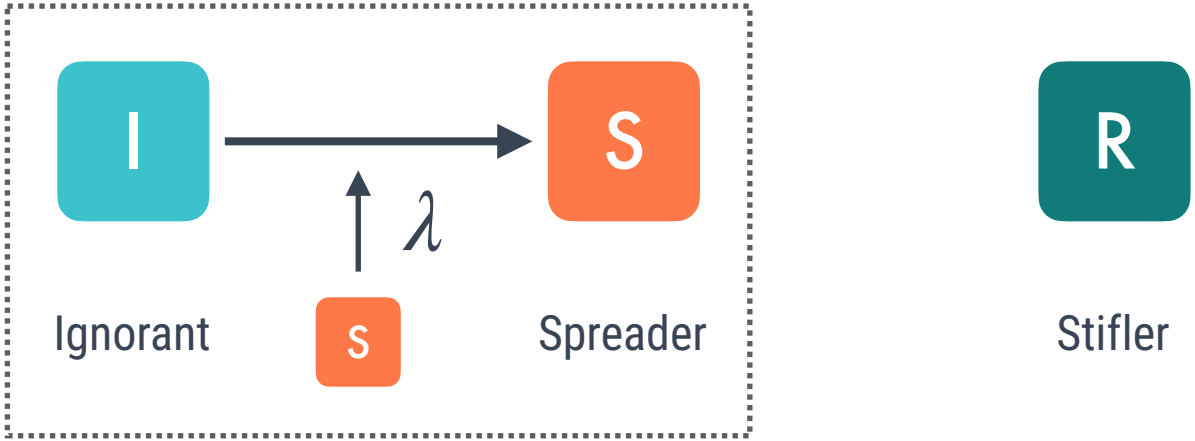
[1] Goffman, W., and Newill, V. A., *Nature*, **204**, 225 (1964).
[2] Daley, D. J., and Kendall, D. G. (unpublished).
[3] Bailey, N. T. J., *The Mathematical Theory of Epidemics* (Griffin, 1960).

Rumor Model



λ :spreading rate

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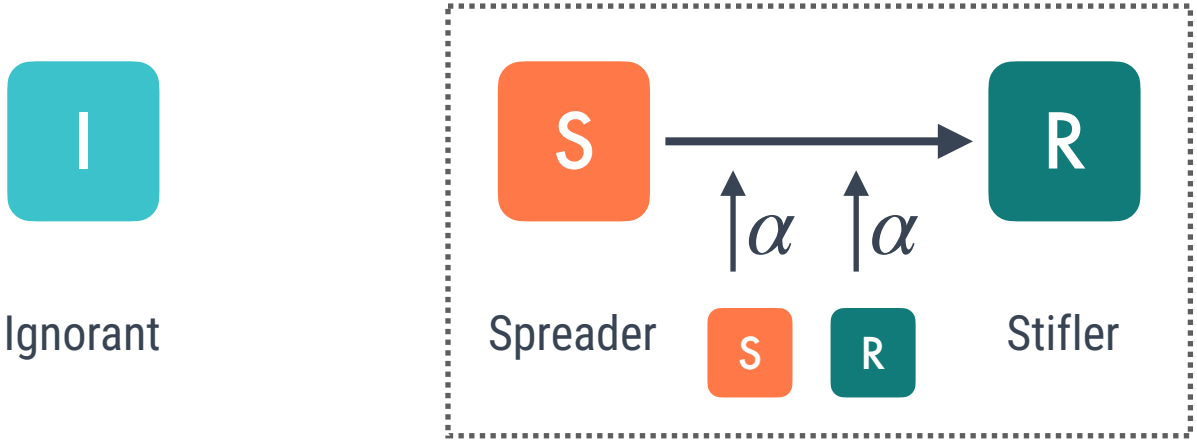
Rumor Model



α :stifling rate

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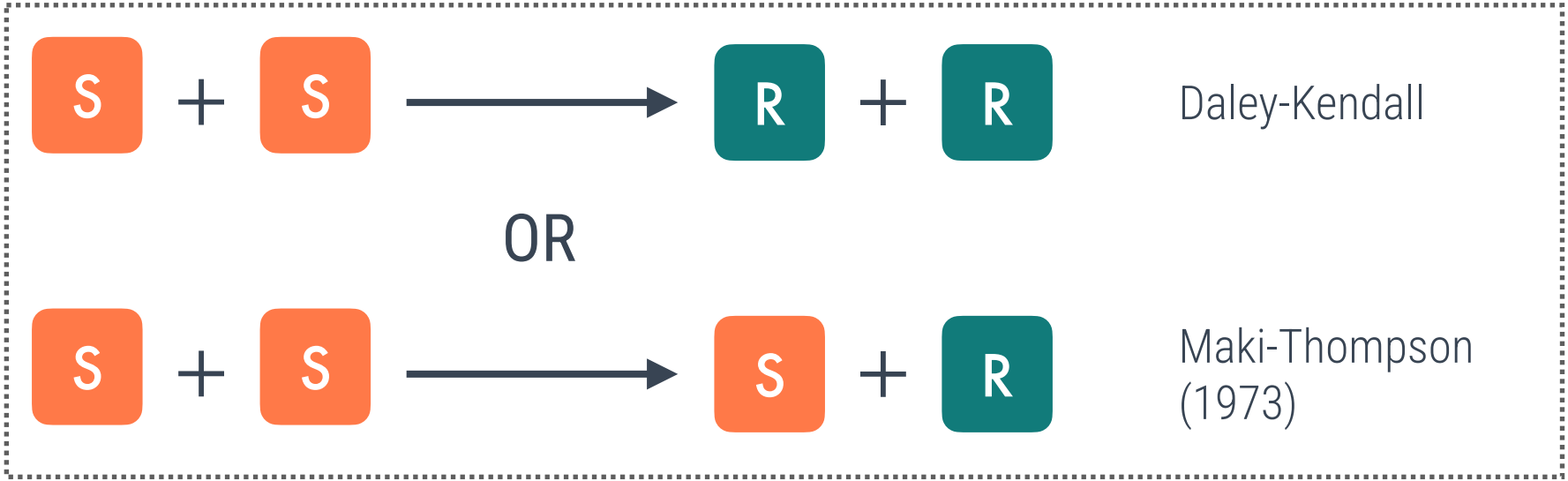
Two “Rumor Models”



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Simple or Complex Contagion?

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What are the deterministic equations?

**Assume Homogeneous Mixing

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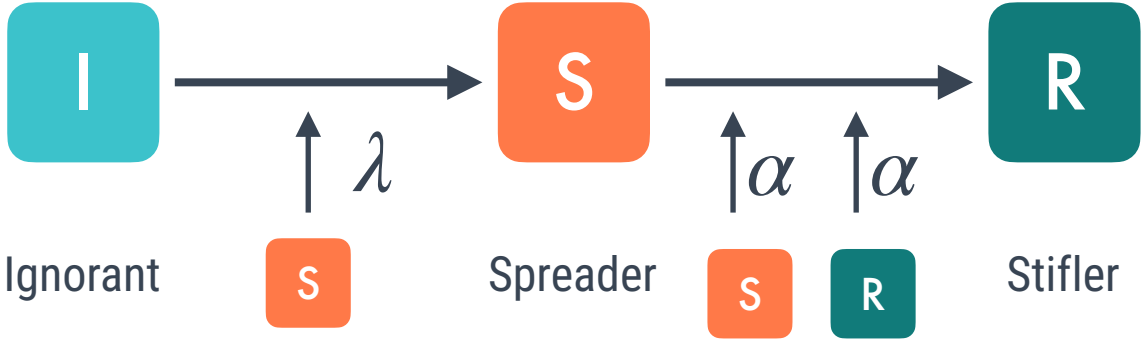
Rumor Model



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Deterministic Equations

$$I(t+1) = I(t) - \lambda \frac{S(t)I(t)}{N}$$

$$S(t+1) = S(t) + \lambda \frac{S(t)I(t)}{N} - \alpha S(t) \frac{(S(t) + R(t))}{N}$$

$$R(t+1) = R(t) + \alpha S(t) \frac{(S(t) + R(t))}{N}$$

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**Still in the Homogeneous Mixing Scenario

What is the force of infection for the rumor model?

1. What is the probability an ignorant will become a spreader at some time t? (I(t), S(t), R(t))
2. What is a the probability a spreader will transition into a stifler at time t? (I(t), S(t), R(t))

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Transition Probabilities

$$P_{I \to S} = 1 - (1 - \frac{\lambda}{N})^{S(t)}$$

$$P_{S \to R} = 1 - (1 - \frac{\alpha}{N})^{(S(t)+R(t))}$$

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What is the threshold condition?

**Assume Homogeneous Mixing

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'Reluctance to tell stale news' can be incorporated into the model in other ways, some of which we have investigated by similar methods, and the results are then correspondingly modified. However, the object of this communication is not to urge the appropriateness of any one model of rumour-telling; but to stress the danger of transferring formulæ designed for epidemiological use without a thorough re-examination of the hypotheses, which may not always be appropriate in the new situation.

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**Assume Homogeneous Mixing

Model Extension: What happens if we forget the rumor?

How would you write the new model?
Deterministic Equations?
Is there a threshold condition?

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What are the deterministic degree block equations?

Threshold condition?

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In-class/ Homework (for next Monday)

- Code up the Rumor Model
    - deterministic/stochastic homogeneous mixing
    - network model
    - plots: final size vs. lambda
    - choose a network property to explore

    Thursday: Assignment!

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